

James Holehouse

✉ jamesholehouse@santafe.edu, jamesholehouse1@gmail.com

🌐 My Website

🐙 GitHub

🎓 Scholar

🌐 LinkedIn

Personal

- I'm a mathematical biologist and complexity scientist at the Santa Fe Institute (SFI). From Fall 2026, I will be an assistant professor in the school of biology at Washington University, St. Louis. I am a British citizen, and I currently work in the USA on a J1 visa. Currently, I am interested in:

- Understanding the role of randomness in gene expression.
- The interplay between stochasticity and non-equilibrium transient effects.
- The evolution of rules and regulations across biological and social systems.

I believe that there are unifying principles governing rules and regulations in biological and social systems, and that an understanding of regulatory function in social systems will help us understand regulatory function at all levels of biology, from gene regulatory networks governing protein-DNA interactions to ecological networks governing the interactions between organisms.

Employment History

Starting Fall 2026	■ Assistant Professor at Washington University in St. Louis Department of Biology.
October 2022 – Present	■ Postdoc at the Santa Fe Institute. Working alongside Sidney Redner, Chris Kempes, Geoffrey West, Vicky Chuqiao Yang and Hyejin Youn. • From 2023-2025 I was the Curriculum Director of the Undergraduate Complexity Researcher program (an REU program). • In both 2023 and 2024 I organized the annual SFI postdoc retreat. • Co-organized a series of talks at St. Johns College from SFI postdocs. • I was a core member of the SFI postdoc selection process for 2023.
May 2021 – October 2021	■ Economic Modeling Intern at Cambridge Econometrics. An industrial internship in complexity economics alongside Hector Pollitt.
Feb 2021 – Feb 2022	■ Tutor at MyTutor. I tutored GCSE and A-level Mathematics and Physics to high school students during the pandemic.
Sept 2019 – June 2021	■ Teaching Assistant at the University of Edinburgh. I taught Python programming to 1st year Biology students.
June 2017 – Aug 2017	■ NERC Summer Researcher at the University of Edinburgh. A climate science summer internship supervised by Simon Tett.
pre 2017	■ I worked at various positions in hospitality including, working at a hostel and as a cook in a restaurant, to support my studies.

Education

- 2018 – 2022  **PhD, the University of Edinburgh** in *Mathematical Biology*.
Studying stochastic models of gene expression with Prof. Ramon Grima. I organized and initiated *stochastic problem solving days* and groups retreats in our research group.
I received the Reinhart Heinrich award from the European Society for Mathematical and Theoretical Biology (2022) for the best thesis in Mathematical Biology.
- 2021  **TensorFlow Certificates** in *Coursera*.
Achieved 99% in courses on *Introduction to TensorFlow for Artificial Intelligence, Machine Learning, and Deep Learning* and *Convolutional Neural Networks in TensorFlow*.
- 2020 – 2021  **Centre for Open Learning, the University of Edinburgh** *Introduction to Philosophy*.
I completed a night-school Philosophy course during my PhD studies.
- 2014 – 2018  **MPhys, the University of Edinburgh** in *Theoretical Physics*.
I achieved a First Class Honors degree (1:1) and did my Masters project alongside Prof. Richard Blythe. My course choices were tailored to mathematical and statistical physics modules.
- 2012 – 2014  **A-Levels, Scarborough Sixth Form College**.
Obtained A*A*A in Mathematics, Physics and Chemistry respectively (A* is top grade), and AS-Levels AA in Further Mathematics and Biology (A is top grade).

Publications

Published Articles

- 1 **Holehouse, J.** (2024). Model reduction, mechanistic modelling and transience in models of stochastic chemical kinetics. *European Communications in Mathematical and Theoretical Biology*, 26, 6–14.
- 2 **Holehouse, J., & Redner, S.** (2024). First passage on disordered intervals. *Physical Review E*, 109(3), L032102. **Corresponding author paper.**
- 3 Wu, B., **Holehouse, J.**, Grima, R., & Jia, C. (2024). Solving the time-dependent protein distributions for autoregulated bursty gene expression using spectral decomposition. *The Journal of Chemical Physics*, 160(7).
- 4 **Holehouse, J.** (2023). Recurrence and eigenfunction methods for non-trivial models of discrete binary choice. *Entropy*, 25(7), 996.
- 5 Weidemann, D. E., **Holehouse, J.**, Singh, A., Grima, R., & Hauf, S. (2023). The minimal intrinsic stochasticity of constitutively expressed eukaryotic genes is sub-poissonian. *Science Advances*, 9(32), eadh5138.
- 6 **Holehouse, J., & Moran, J.** (2022). Exact time-dependent dynamics of discrete binary choice models. *Journal of Physics: Complexity*. **Corresponding author paper.**
- 7 **Holehouse, J., & Pollitt, H.** (2022). Non-equilibrium time-dependent solution to discrete choice with social interactions. *PLoS ONE*. **Corresponding author paper.**
- 8 Braichenko, S., **Holehouse, J.**, & Grima, R. (2021). Distinguishing between models of mammalian gene expression: Telegraph-like models versus mechanistic models. *Journal of the Royal Society Interface*. **Joint first author publication.**
- 9 **Holehouse, J.**, Cao, Z., & Grima, R. (2020). Stochastic modeling of autoregulatory genetic feedback loops: A review and comparative study. *Biophysical journal*, 118(7), 1517–1525.
- 10 **Holehouse, J.**, Gupta, A., & Grima, R. (2020). Steady-state fluctuations of a genetic feedback loop with fluctuating rate parameters using the unified colored noise approximation. *Journal of Physics A: Mathematical and Theoretical*, 53(40), 405601.

- 11 **Holehouse, J.,** Sukys, A., & Grima, R. (2020). Stochastic time-dependent enzyme kinetics: Closed-form solution and transient bimodality. *The Journal of Chemical Physics*, 153(16), 164113.
- 12 **Holehouse, J.,** & Grima, R. (2019). Revisiting the reduction of stochastic models of genetic feedback loops with fast promoter switching. *Biophysical journal*, 117(7), 1311–1330.
- 13 **Holehouse, J.,** & Cameron, B. (2017). Meteorological visibility observations: A user's guide. *Times Educational Supplement*.

Other and In Progress

- 1 **Holehouse, J.** (2025a). Do Distinct Subpopulations Signify Modes of Behavior in a Noisy Single Cell? (Submitted).
- 2 **Holehouse, J.** (2025b). From Single Genes to Populations: Quantifying Broken Detailed Balance in Transcription. arXiv preprint arXiv:2405.12897.
- 3 **Holehouse, J.,** Redner, V. C., S. and Yang, Krapivsky, H., P.L. and Youn, Arroyo, J. I., Redner, S., West, G. B., & Kempes, C. P. (2025). Unifying function diversity patterns across biological and social systems. Working paper.
- 4 **Holehouse, J.,** Yang, V. C., Kempes, C. P., Youn, H., Arroyo, J. I., Redner, S., & West, G. B. (2025). Scaling Laws for Function Diversity and Specialization Across Socioeconomic and Biological Complex Systems. arXiv preprint arXiv:2208.06487.
- 5 **Holehouse, J.** (2022). Model reduction, mechanistic modelling and transience in models of stochastic chemical kinetics. PhD thesis. *The University of Edinburgh*.
- 6 **Holehouse, J.,** & Blythe, R. A. (2018). Cross-situational learning of large lexicons with finite memory. arXiv preprint arXiv:1809.11047.

Seminars, Posters and Invited Talks

- 2025
 - **Seminar** at the UPenn's *MathBio Seminar Series*.
Regulation, Organization and Stochasticity across the Tree of Life.
 - **Workshop Talk** at the Santa Fe Institute's *Emergent Engineering Technology Summit*.
Leveling the playing field: Rule evolution in sports, governance and biology.
 - **Seminar** at Brown University's Data Science Institute.
Rules, Regulations and Randomness: From Biological to Social Systems.
 - **Seminar** at UC Berkeley's Department of Biology.
Rules, Regulations and Randomness: From Biological to Social Systems.
 - **Seminar** at WashU's Department of Biology.
Rules, Regulations and Randomness: From Biological to Social Systems.
- 2024
 - **Workshop talk** at the Santa Fe Institute's *Emergent Engineering Workshop*.
Unnecessary Bureaucracy? Understanding the growth of rules and unifying principles in biological and social systems.
 - **Invited talk** at the London Mathematical Laboratory's Ergodicty Economics seminar series.
First-passage time distributions in one-dimensional ruin problems.
 - **Invited talk** at the University of Warwick, Departments of Mathematics and Life Sciences.
Universal principles of regulation: diversity, stochasticity, and constraints.
 - **Seminar** at the University of Warwick, Departments of Mathematics and Life Sciences.
Universal principles of regulation: diversity, stochasticity, and constraints.
 - **Invited talk** at the Postdocs in Complexity: Global Summit, Santa Fe Institute, USA.
Population dynamics of viral cheating with length-dependent selection.

Seminars, Posters and Invited Talks (continued)

- **Contributed talk** at the Conference on Complex Systems 2024 in Exeter, UK.
First passage on disordered intervals.
- **Plenary talk** at the European Conference for Mathematical and Theoretical Biology 2024 in Toledo, Spain.
Breaking steady state assumptions in models of stochastic chemical kinetics.
- **Contributed talk** at the European Conference for Mathematical and Theoretical Biology 2024 in Toledo, Spain.
The statistical physics of transcription and mRNA expression heterogeneity.
- **Contributed talk** at the 10th International Conference on Computational Social Science (IC²S²) in Philadelphia, USA.
Scaling of diversity and specialization across socioeconomic and biological complex systems.
- **Invited talk** at the University of California Berkeley at the lab of Prof. Hernan Garcia.
What are the benefits of bursty transcription?
- **Invited talk** at St. Johns College, Santa Fe, USA.
A complexity approach to rules, regulations and bureaucracy. This is part of the ongoing seminar series I am co-organizing.
- 2023 ■ **Workshop talk** at the Santa Fe Institute, USA.
A unified mechanistic model of diversity in cities, cells and companies.
- 2022 ■ **Poster presentation** at the European Conference for Mathematical and Theoretical Biology 2022, Heidelberg.
Analytics and Bimodality in Transient Michaelis-Menten Kinetics.
- **Seminar** at Cambridge Econometrics, UK.
Stochastic methods for binary decision models.
- **Seminar** at Scuola Superiore Sant'Anna, Italy.
Stochastic methods for binary decision models.
- **Invited talk** at the Santa Fe Institute, USA.
Time-dependent solutions to master equations in chemical kinetics and opinion formation.
- 2021 ■ **Seminar** at the University of St. Andrews, UK.
Stochastic time-dependent enzyme kinetics: Closed-form solution and transient bimodality.
- 2019 ■ **Invited talk** at the University of Edinburgh, UK.
Using moment-based maximum likelihood inference to infer parameters from experimental data.

Mentoring

- 2025 ■ **Leonardo Ivan Estrella (Minerva University).**
Intrinsic and extrinsic noise in the evolution of gene regulatory networks.
- **Daniel Chia (Carnegie Melon University).**
The evolution of rules in cricket and rugby.
- 2024 ■ **Edwina Martinez (Santa Fe Indian School).**
Increasing the complexity of cellular automata.
- **Elena Parkerson (Carleton College)** at the Santa Fe Institute.
Finite state projection and stochastic simulation for viral infections.
- **Shloka Janapaty (Colombia University)** at the Santa Fe Institute.
Reaction-diffusion equations of biocrust growth.

Mentoring (continued)

- 2023-present  **Anish Pandya (The University of Texas at Austin)** at the Santa Fe Institute.
Volume-dependence in stochastic gene expression (2023)—presented at the Bulletin of the American Physical Society in Austin, TX.
Causation does not imply correlation: non-intuitive insights from a simple genetic mechanism (2024).
- 2023  **Nathan Hasegawa (Harvey Mudd College)** at the Santa Fe Institute.
Island growth models with preferential attachment.

Grants

- 2025  **NSF PD 22-1332 Social Psychology** grant for **\$262,156** awarded to Andrew Stier (PI, Santa Fe Institute), James Holehouse (co-PI, Santa Fe Institute), and Marc Berman (co-PI, University of Chicago).
Collaborative Research: Understanding attitude formation at different spatial scales.
- 2024  **SFI working group** for **\$4,536**. Participants: Augustinas Sukys and Kaan Öcal.
On the origins of transient bimodality.
-  **SFI working groups** for **\$6,104** and **\$6,690** alongside Asher Leeks (Yale University). Participants: Brandon Schlomann and William DeWitt.
Modeling the evolutionary dynamics of cheating in natural viral infections.
- 2023  **Lou Schuyler Grant** for **\$14,640** from the Santa Fe Institute.
Understanding the trade-offs between diversity, robustness and performance across the tree of life.
- 2020  **Scottish Mathematical Biology Forum (SMBF) 2020 Research Collaboration Prize** for **£500** alongside Jochen Kursawe (University of St. Andrews).
The physical meaning and application of Hill functions in gene regulatory networks.

Service and Skills

Journal Reviewing

- 2020–present  **Reviewer** for Mathematical Biosciences.
- 2021–present  **Reviewer** for PLoS One.
-  **Reviewer** for Physical Review E.
- 2023–present  **Reviewer** for Physical Review Letters.
-  **Reviewer** for Biophysical Journal.
- 2024–present  **Reviewer** for Chaos: An Interdisciplinary Journal of Nonlinear Science.
-  **Reviewer** for npj Complexity.
-  **Reviewer** for the Journal of Statistical Mechanics: Theory and Experiment.

Programming Languages

- Julia  Very proficient, preferred language.
- Mathematica  Very proficient, preferred language.
- Python  Very proficient.